

Kushagra Tiwari

+91 8318661731 | Mail | LinkedIn | GitHub | Portfolio

EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal

Bachelor of Technology in Electrical and Electronics Engineering

Aug 2023 – Present

CGPA: 7.14/10

EXPERIENCE

Magi

Apr 2026 – Present

Software Development Engineer Intern

Remote

- Built an **event-driven scheduler** for recurring jobs, delayed execution, distributed automation, contextual dispatch, and retries.
- Integrated scheduler execution with a workflow **orchestrator** for task chaining, workflow state handling, and agent-triggered jobs.
- Designed backend APIs for **AI generation workflows**, competitive intelligence, content pipelines, job lifecycle state, and automation endpoints.
- Exposed workflows as **MCP tools**, enabling agents to discover APIs, invoke actions, validate inputs, and execute backend capabilities.

PipesHub

Jan 2026 – Jun 2026

Software Development Engineer Intern

Remote

- Built production-grade **SaaS connectors** for AI search, agents, workflow automation, enterprise integrations, and tool execution.
- Engineered a secure **Electron bridge** for filesystem access, sandboxed execution, host orchestration, and desktop-to-cloud workflows.
- Designed recursive **file sync pipelines** with delta tracking, resumability, conflict-safe writes, failure recovery, and metadata handling.
- Implemented schema-driven **LLM action validation** for runtime safety, deterministic tool execution, input validation, and predictable agents.

Acredge

Nov 2025 – Apr 2026

Software Development Engineer Intern

Remote

- Reduced **retrieval latency by 60%** by parallelizing search pipelines, optimizing ranking stages, and removing bottlenecks.
- Built an **AI orchestration system** for real-estate workflows using n8n, LLM APIs, ranking services, automation tools, and webhooks.
- Implemented **Recombee pipelines** for relevance ranking, personalization, behavioral recommendations, and retrieval quality for recommendation system.
- Resolved production bottlenecks across retrieval, orchestration, recommendations, workflow reliability, and API response paths.

LeastActions

Oct 2025 – Feb 2026

Software Development Engineer Intern

Remote

- Built backend APIs for **sandboxed code execution**, collaborative IDE workflows, workspace state, execution metadata, and job results.
- Designed modular API contracts, validation layers, state handling, and request flows to improve stability and reduce regressions.
- Deployed services on **AWS EC2** with CI/CD, Linux setup, Nginx configuration, reverse proxying, and production process management.
- Improved backend reliability by refining APIs, simplifying deployments, hardening infrastructure workflows, and debugging runtime failures.

PROJECTS

🔗 Gapi | Go, net/http, OpenAPI 3.1, CLI, Validation, DI

Jun 2026 – Jul 2026

- Authored a **FastAPI-inspired Go framework** with typed generic handlers, route helpers, middleware, groups, and idiomatic **net/http**.
- Implemented request binding for path, query, header, cookie, body, validation tags, custom validators, and type-safe handler signatures.
- Built **OpenAPI 3.1 generation**, docs UIs, RFC 9457 problem details, response serialization, security metadata, testing helpers, and CLI tooling.

🔗 VaultLens | Electron, TypeScript, SQLite, RAG, Embeddings

May 2026 – Jun 2026

- Built a **local-first AI memory engine** using SQLite, vector embeddings, semantic retrieval, private desktop indexing, and local storage.
- Designed **RAG indexing pipelines** with chunking, metadata extraction, optimized retrieval, evidence-grounded generation, and citations.
- Implemented retrieval tracing, task extraction, timeline reconstruction, context-aware synthesis, and actionable memory workflows.

TECHNICAL SKILLS

Languages

TypeScript, JavaScript, Python, Go, SQL, C++, Bash

Backend

Node.js, NestJS, Express.js, FastAPI, Django, REST APIs, WebSockets, JWT, RBAC

AI

RAG, AI Agents, MCP, LangChain, LangGraph, Tool Calling, Embeddings, Vector Search, LLM APIs

Systems

Redis, Kafka, Celery, BullMQ, Async Workers, Job Queues, Cron Schedulers, Event-Driven Systems

Databases

PostgreSQL, MongoDB, SQLite, Redis, Firebase, Vector DBs, SQL Optimization, Schema Design

DevOps

Docker, AWS EC2, Linux, Nginx, GitHub Actions, CI/CD, Reverse Proxying

Frontend

React, Next.js, Electron, React Native, Flutter, Swift, Tailwind CSS